

Top open source quantum computing frameworks

Open source frameworks provide the necessary interface, tools, and resources for conceptualising, simulating, and testing quantum algorithms across many different hardware platforms. Each framework embodies a different philosophy — some provide gate-based programming, while others focus on annealing or photonic frameworks. Here's a quick look at how the leading open source frameworks work and what they are good for.

Qiskit

One of the most well-known quantum SDKs is IBM's Qiskit. Built in Python, it has a modular design with the following libraries.

- *Terra*: Allows building and compiling quantum circuits.
- *Aer*: Simulates circuits using classical hardware.
- *Aqua*: Application-level algorithms for chemistry, AI, and optimisation.

Qiskit interfaces directly with the IBM Quantum Experience allowing users to conduct experiments with quantum processors hosted in the cloud. It also has a very wide ecosystem with a highly active worldwide community, making it the most adopted framework today.

Use cases of Qiskit

- *Quantum chemistry and drug discovery*: Accelerating molecular simulations in drug research and innovation.
- *Portfolio optimisation in finance*: Better portfolio allocation in times of uncertainty.
- *Quantum-enhanced machine learning*: Accelerating classification, clustering and pattern recognition.
- *Optimising supply chain and logistics*: Reducing the operational inefficiencies in supply chain and logistics.
- *Cryptography and security research*: Experimentation with post-quantum

cryptographic protocols.

- *Innovation in materials science*: Simulating properties of advanced materials for renewable energy and electronics.

Cirq

Written in Python, Google's Cirq is an open source framework designed for building, simulating, and running quantum circuits on Noisy Intermediate-Scale Quantum (NISQ) machines. Cirq is not a general-purpose SDK but is designed to be anchored around a specific device (Google's Sycamore processor). It provides simulation for algorithms running in a realistic hardware environment, and offers support for hybrid quantum-classical modes of operation.

Cirq allows researchers to work on increasingly sophisticated quantum error correction prototypes, benchmarking the performance of quantum hardware and working with novel quantum algorithms.

Use cases of Cirq

- *Simulations of quantum circuits*: Modelling circuits, which gives us an insight into how they run on noisy processors.
- *Error mitigation*: Developing strategies to reduce errors in NISQ devices.
- *Variational quantum algorithms*: Hybrid approaches that allow us to extend the application of optimisation, chemistry, and AI.
- *Benchmarking NISQ hardware*: Testing the fidelity of devices and evaluating the performance of processors against one another.
- *Experimenting with quantum machine learning*: Prototyping ML models together with real world data and unleashing quantum features.

PennyLane (Xanadu)

Developed by Xanadu, PennyLane is perhaps the best known open source quantum machine learning (QML) library. This hybrid quantum-classical

computing library is focused on the developing field of QML and is built entirely in Python. It has seamless compatibility with some of the most widely used deep learning frameworks such as TensorFlow, PyTorch and JAX, enabling researchers and developers to easily train quantum circuits with classical neural networks.

The main feature of PennyLane pertaining to quantum AI has to do with differentiable programming, which allows for the automatic differentiation of quantum circuits. It is capable of exploratory experiments and deploying virtually any quantum hardware.

Use cases of PennyLane

- *Quantum neural networks (QNNs)*: Adding quantum layers to a classical deep learning architecture.
- *Variational quantum algorithms (VQAs)*: Making solving eigenvalue problems or optimisation problems more feasible.
- *Differentiable programming*: Enabling end-to-end optimisation.
- *Quantum chemistry simulations*: Generating molecular property estimates and expediting computational chemistry.
- *Quantum-enhanced generative models*: Used for investigating quantum enhancements in generative adversarial networks and probabilistic modelling.
- *Education and research*: Many universities use PennyLane for research on quantum AI. Because of its focus on hybrid architectures, PennyLane will play a significant role in future quantum AI applications.

Forest SDK

Rigetti's Forest SDK is a framework for quantum development and research built in Python. It has two components:

- *pyQuil*: A library written in Python that allows for quantum program construction, simulation and execution.